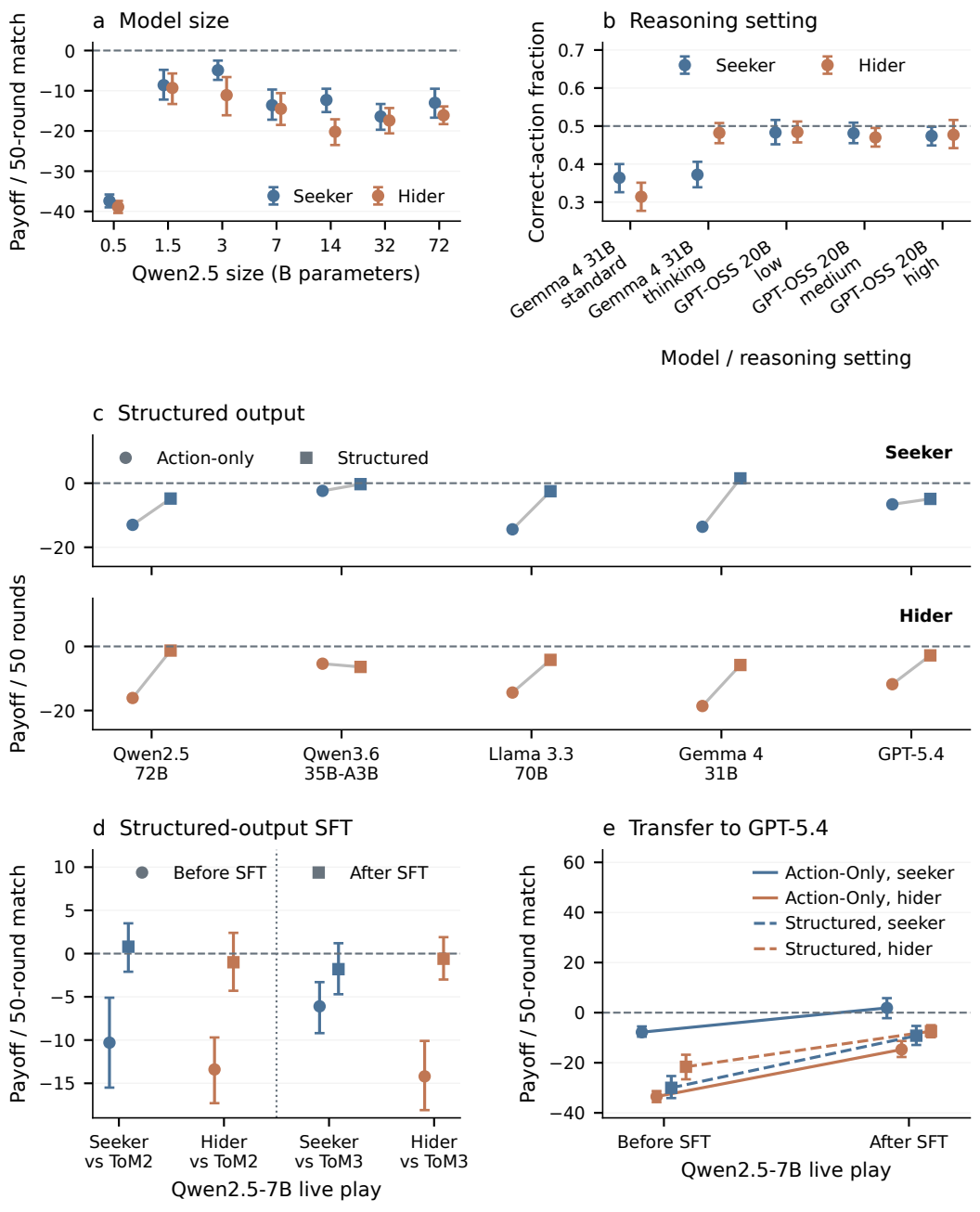
a Model size
Payoff / 50-round match
0
−10
−20
−30
−40
Seeker
Hider
0.5
1.5
3
7
14
32
72
Qwen2.5 size (B parameters)
b Reasoning setting
Correct-action fraction
0.7
0.6
0.5
0.4
0.3
Seeker
Hider
Gemma 4 31B standard
Gemma 4 31B thinking
GPT-OSS 20B low
GPT-OSS 20B medium
GPT-OSS 20B high
Model / reasoning setting
c Structured output
Action-only
Structured
Seeker
Hider
Payoff / 50 rounds
0
−20
Qwen2.5 72B
Qwen3.6 35B-A3B
Llama 3.3 70B
Gemma 4 31B
GPT-5.4
d Structured-output SFT
Payoff / 50-round match
10
5
0
−5
−10
−15
Before SFT
After SFT
Seeker vs ToM2
Hider vs ToM2
Seeker vs ToM3
Hider vs ToM3
Qwen2.5-7B live play
e Transfer to GPT-5.4
Payoff / 50-round match
60
40
20
0
−20
−40
Action-Only, seeker
Action-Only, hider
Structured, seeker
Structured, hider
Before SFT
After SFT
Qwen2.5-7B live play